Bayesian Optimization Report: Function 1

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.736	Good
2	0.709	Good
3	0.716	Good
4	0.714	Good
5	0.707	Good
6	0.702	Good
7	0.696	Medium
8	0.698	Medium
9	0.701	Good
10	0.703	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.0005	0.0001	0.0008
2	0.0003	0.0000	0.0008
3	0.0003	0.0000	0.0009
4	0.0003	0.0000	0.0008
5	0.0003	0.0000	0.0009
6	0.0003	0.0000	0.0008
7	0.0003	0.0000	0.0008
8	0.0003	0.0000	0.0008
9	0.0003	0.0000	0.0008
10	0.0002	0.0000	0.0009

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies

1	[0.92779382 0.68128984]	-0.0035656083250481386	Matern(ν=0.5)	EI	0.736	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.9226056997161378)}}
2	[1. 0.67719893]	-0.0036168883595787066	Matern(ν=0.5)	EI	0.709	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.7393336576288002)}}
3	[0.87715729 0.6769563]	-0.0036194149683447087	Matern(ν=1.5)	EI	0.716	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.7501419173672943)}}
4	[0.91060791 0.67660891]	-0.0036148508918724645	Matern(ν=0.5)	EI	0.714	None
5	[1. 0.67252352]	-0.0035513989868064108	Matern(ν=2.5)	EI	0.707	None
6	[0.87474045 0.67735752]	-0.0036194979995481796	Matern(ν=0.5)	EI	0.702	None
7	[1. 0.67317964]	-0.00356815583 7315503	Matern(ν=0.5)	EI	0.696	None
8	[0.89272583 0.67385348]	-0.00358386453 0960306	Matern(ν=0.5)	EI	0.698	None
9	[0.55111521 0.67479405]	-0.0036196796935227676	Matern(ν=0.5)	EI	0.701	None
10	[0.97870562 0.67623782]	-0.0036142604210674025	Matern(ν=2.5)	EI	0.703	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.001794
EI	0.000000
PI	0.000000
THOMPSON	0.001356
KG	0.000000
ENTROPY	0.675981
PORTFOLIO	0.002406

Iteration 2

Method	Max Value
UCB	0.000587
EI	0.000000
PI	0.000000
THOMPSON	0.000817
KG	0.000000
ENTROPY	0.651437
PORTFOLIO	0.002111

Iteration 3

Method	Max Value
UCB	0.000532
EI	0.000000
PI	0.000000
THOMPSON	0.001456
KG	0.000000
ENTROPY	0.657934
PORTFOLIO	0.002102

Iteration 4

Method	Max Value
UCB	0.000568
EI	0.000000
PI	0.000000
THOMPSON	0.000950
KG	0.000000
ENTROPY	0.655759
PORTFOLIO	0.001757

Iteration 5

Method	Max Value
UCB	0.000474
EI	0.000000
PI	0.000000

THOMPSON	0.001230
KG	0.000000
ENTROPY	0.649547
PORTFOLIO	0.001839

Iteration 6

Method	Max Value
UCB	0.000545
EI	0.000000
PI	0.000000
THOMPSON	0.001284
KG	0.000000
ENTROPY	0.644986
PORTFOLIO	0.001618

Iteration 7

Method	Max Value
UCB	0.000541
EI	0.000000
PI	0.000000
THOMPSON	0.001358
KG	0.000000
ENTROPY	0.640009
PORTFOLIO	0.001572

Iteration 8

Method	Max Value
UCB	0.000536
EI	0.000000
PI	0.000000
THOMPSON	0.001187
KG	0.000000
ENTROPY	0.641238
PORTFOLIO	0.001463

Iteration 9

Method	Max Value
UCB	0.000536
EI	0.000000
PI	0.000000
THOMPSON	0.000841
KG	0.000000
ENTROPY	0.643763
PORTFOLIO	0.001410

Iteration 10

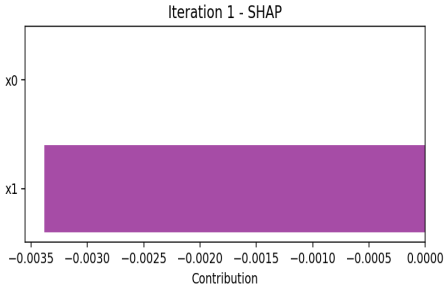
Method	Max Value
UCB	0.000570
EI	0.000000
PI	0.000000
THOMPSON	0.000845
KG	0.000000
ENTROPY	0.645879
PORTFOLIO	0.001472

LOO MSE Summary

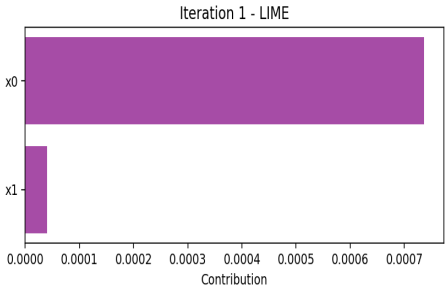
Iteration	LOO MSE
1	0.000001
2	0.000000
3	0.000000
4	0.000000
5	0.000000
6	0.000000
7	0.000000
8	0.000000
9	0.000000
10	0.000000

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



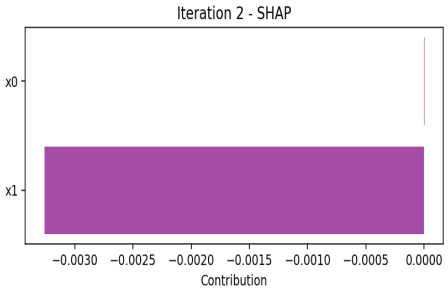
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-0.00000
x1	-0.00338

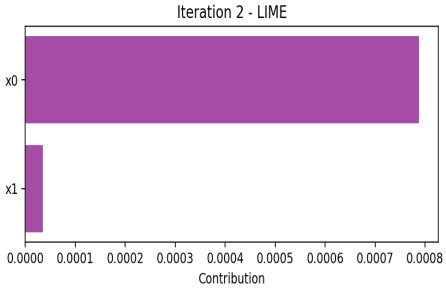
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.00074
x1	0.00004

Iteration 2 - SHAP



Iteration 2 - LIME



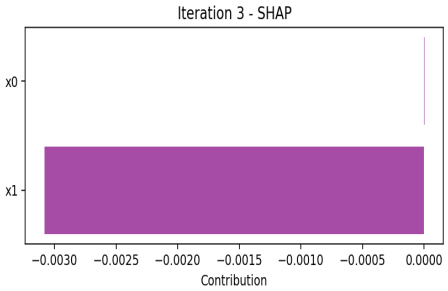
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00326

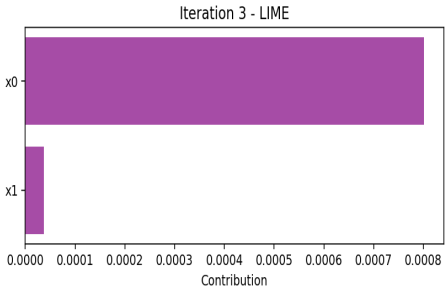
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.00079
x1	0.00003

Iteration 3 - SHAP



Iteration 3 - LIME



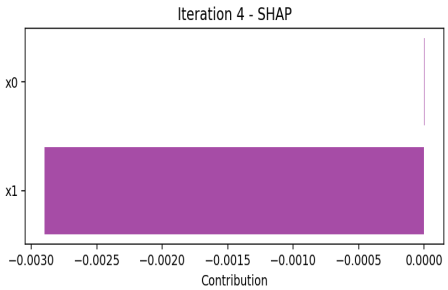
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00308

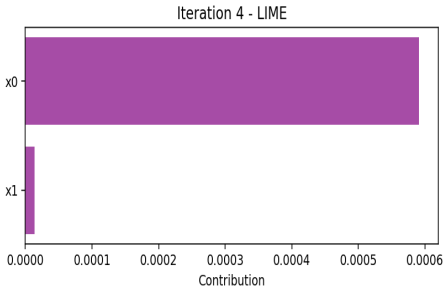
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.00080
x1	0.00004

Iteration 4 - SHAP



Iteration 4 - LIME



Iteration 4 - SHAP Values

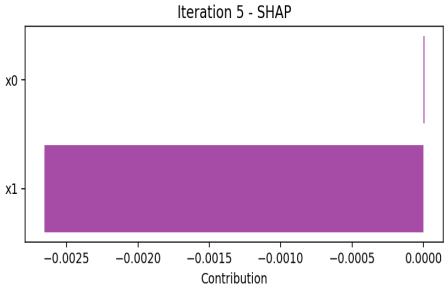
Feature	SHAP Value
x0	0.00001
x1	-0.00290

Iteration 4 - LIME Values

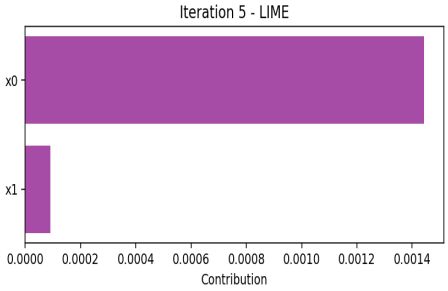
Feature	LIME Value
x0	0.00059

x1	0.00001

Iteration 5 - SHAP



Iteration 5 - LIME



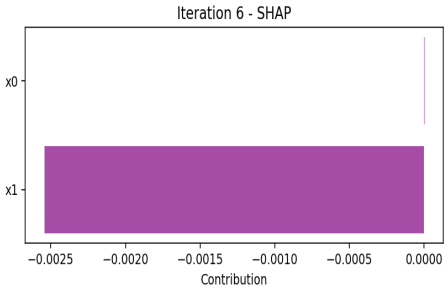
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	-0.00266

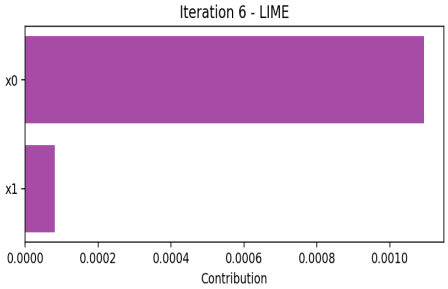
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.00144
x1	0.00009

Iteration 6 - SHAP



Iteration 6 - LIME



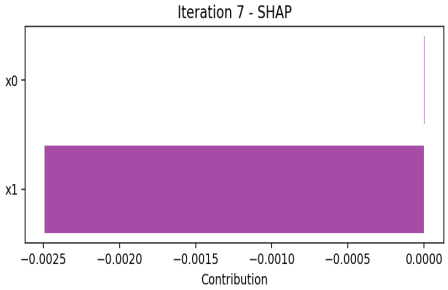
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00254

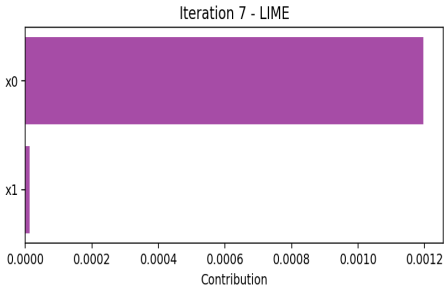
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.00109
x1	0.00008

Iteration 7 - SHAP



Iteration 7 - LIME



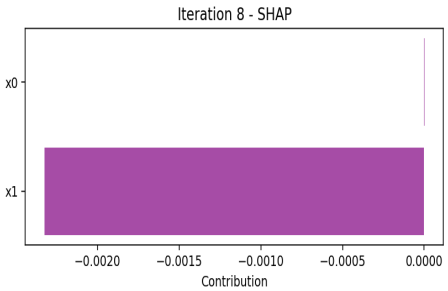
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00249

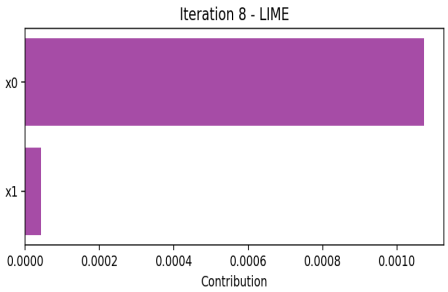
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.00120
x1	0.00001

Iteration 8 - SHAP



Iteration 8 - LIME



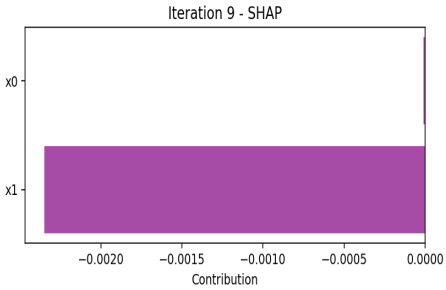
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.00000
x1	-0.00233

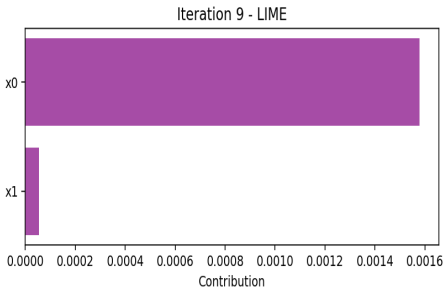
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.00107
x1	0.00004

Iteration 9 - SHAP



Iteration 9 - LIME



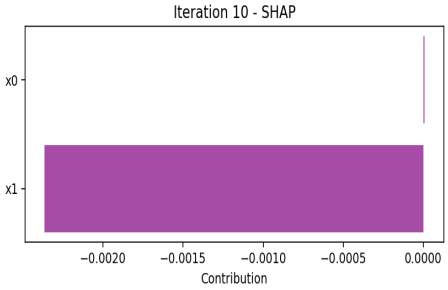
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	-0.00001
x1	-0.00235

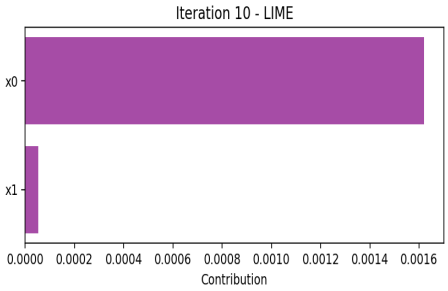
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.00158
x1	0.00006

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.00001
x1	-0.00237

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.00162
x1	0.00005